Lecture 19

Working procedure for dual Simplex method

Step 1
a) Convert to Maximization form
b) Convert ($\geq$) type constraints to ($\leq$) type
c) Standard form.

Step 2 Find initial basic soln. & express in form of dual simplex table

Step 3 $C_j = c_j - z_j$
a) If all $C_j \leq 0$; and all $b_i \geq 0$, then optimal basic feasible soln. has been attained.

b) If all $C_j \leq 0$, and at least one $b_i < 0$, go to step 4
c) If any $C_j \geq 0$, then method fails

Step 4 Mark out going variable
↳ row that contains most negative b_i.

Step 4 Test nature of key row elements
a) if all elements are ≥ 0, problem has no feasible soln.
b) if at least one element < 0; find ratios of corresponding elements of C_j - row to these elements
choose the smaller ratio & column is key column

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⑥ Iterate toward optimal feasible soln

Ex Use dual simplex method

Max. $-3x_1 - 2x_2$

s.t.
$$x_1 + x_2 \geq 1$$
$$x_1 + x_2 \leq 7$$
$$x_1 + 2x_2 \geq 10$$
$$x_2 \leq 3$$
$$x_1 \geq 0, x_2 \geq 0.$$

$$x_1 + x_2 \geq 1 \longrightarrow -x_1 - x_2 \leq -1$$
$$x_1 + 2x_2 \geq 10 \longrightarrow -x_1 - 2x_2 \leq 10$$

Standard form

Max. $Z = -3x_1 - 2x_2 + 0s_1 + 0s_2 + 0s_3 + 0s_4$

s.t.
$$-x_1 - x_2 + s_1 = -1$$
$$x_1 + x_2 + s_2 = 7$$
$$-x_1 - 2x_2 + s_3 = 10$$
$$x_2 + s_4 = 3$$

$x_1, x_2 = 0$; $s_1 = -1$, $s_2 = 7$, $s_3 = 10$, $s_4 = 3$ +

$Z = 0$

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L19-2a

Simplex Table

	C_j	-3	-2	0	0	0	0	
C_B	Basis	x_1	x_2	S_1	S_2	S_3	S_4	b
0	S_1	-1	-1	1	0	0	0	-1
0	S_2	1	1	0	1	0	0	7
0	S_3	-1	$\boxed{-2}$	0	0	1	0	-10 ← ①
0	S_4	0	1	0	0	0	1	3
$Z_j = \Sigma C_B a_{ij}$		0	0	0	0	0	0	
$C_j = C_j - Z_j$		-3	-2	0	0	0	0	
$\theta =$		-3/-1	-2/-2	-	-	1/0	-	

-ve + numerically largest

↑ ② smallest ratio

out going S_3 ; in coming x_2

$R_3' \to R_3/-2$; $R_1' \to R_1 + R_3'$; $R_2' \to R_1 - R_3'$; $R_4' \to R_4 - R_3'$

	C_j	-3	-2	0	0	0	0	
C_B	Basis	x_1	x_2	S_1	S_2	S_3	S_4	b
0	S_1	-1/2	0	1	0	-1/2	0	4
0	S_2	1/2	0	0	1	1/2	0	2
-2	x_2	1/2	1	0	0	-1/2	0	5
0	S_4	$\boxed{-\frac{1}{2}}$	0	0	0	$\frac{1}{2}$	1	-2 ← ①
$Z_j = \Sigma C_B a_{ij}$		-1	-2	0	0	1	0	7
$C_j = C_j - Z_j$		-2	0	0	0	-1	0	
θ		4	0	0	0	-2	0	

↑ ② $\boxed{\text{as +ve}}$

out var S_4 & in var x, $R_4' \to R_4/(-1/2)$

$R_1' \to R_1 + \frac{1}{2}R_4'$, . . .

	C_j	-3	-2	0	0	0	0	b
	Basis	x_1	x_2	S_1	S_2	S_3	S_4	
0	S_1	0	0	1	0	-1	-1	6
0	S_2	0	0	0	1	1	1	0
-2	x_2	0	1	0	0	0	1	3
-3	x_1	1	0	0	0	-10	-2	4
	Z_j	-3	-2	0	0	3	4	
	C_j	0	0	0	0	-3	-4	

all $C_j \le 0$ & all $b \ge 0$,

$\therefore$ this is optimal + feasible

$x_1 = 4$, $x_2 = 3$ & $Z_{max} = -18$

Ex// dual simplex method

Min $Z = 2x_1 + 2x_2 + 4x_3$

s.t. $2x_1 + 3x_2 + 5x_3 \ge 2$

$3x_1 + x_2 + 7x_3 \le 3$

$x_1 + 4x_2 + 6x_3 \le 5$

$x_1, x_2, x_3 \ge 0$

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L19-3a

Standard

$$\text{Max. } Z' = -2x_1 - 2x_2 - 4x_3 + 0s_1 + 0s_2 + 0s_3$$

$$\text{s.t.} \quad -2x_1 - 3x_2 - 5x_3 + s_1 + 0s_2 + 0s_3 = -2$$

$$3x_1 + x_2 + 7x_3 + 0s_1 + s_2 + 0s_3 = 3$$

$$x_1 + 4x_2 + 6x_3 + 0s_1 + 0s_2 + s_3 = 5$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

$x_1 = x_2 = x_3 = 0$ (non basic

$s_1 = -2,\ s_2 = 3,\ s_3 = 5$ (basic) $Z' = 0$

	c_j	-2	-2	-4	0	0	0		
C_B	Basis	x_1	x_2	x_3	s_1	s_2	s_3	b	
0	s_1	-2	-3	-5	1	0	0	-2	← 1
0	s_2	3	1	7	0	1	0	3	
0	s_3	1	4	6	0	0	1	5	
	z_j	0	0	0	0	0	0		
	c_j	-2	-2	-4	0	0	0		
	θ	$-2/-2$	$-2/-3$	$-4/-5$					

↑

2 smaller +ve

c_j

Basis	x_1	x_2	x_3	s_1	s_2	s_3	b
x_2	$2/3$	1	$5/3$	$-1/3$	0	0	$2/3$
s_2	$7/3$	0	$16/3$	$1/3$	1	0	$7/3$
s_3	$-5/3$	0	$-2/3$	$4/3$	0	1	$7/3$
Z_j	$-4/3$	-2	$-10/3$	$2/3$	0	0	$-4/3$
c_j	$-2/3$	0	$-2/3$	$-2/3$	0	0	

all $c_j \leq 0$ & all b_i are > 0.

soln is optimal & feasible

$x_1 = 0$, $x_2 = 2/3$ & $x_3 = 0$ & max $Z' = -4/3$

min. $Z = 4/3$

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L19-4a

Transportation problem

Special class of LPP in which objective is to transport a single commodity from various origins to different destination at a minimum cost.

There are m plant locations or warehouse (origin) and n distribution centres (destination).

Production capacity of ith plant is a_i

No. of unit required at j^{th} destination is b_j.

Transportation cost of one unit from i^{th} plant to j^{th} destination $\rightarrow$ C_{ij}.

Objective is to determine no. of units to be transported from i^{th} plant to j^{th} destination so that total transportation cost is minimum.

$x_{ij} \rightarrow$ no. of units shipped from i^{th} plant to j^{th} destination

One can write general transportation problem as.

$$\text{Min. } Z = \sum_{i=1}^{m} \sum_{j=1}^{n} C_{ij} x_{ij}$$

s.t.

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$x_{i1} + x_{i2} + \ldots + x_{in} = a_i \rightarrow$
for i^{th} origin $(i = 1, 2, \ldots, m)$

$x_{1j} + x_{2j} + \ldots + x_{mj} = b_j$
for j^{th} destination $(j = 1, 2, \ldots, n)$

physical so $x_{ij} \geq 0$

$\Rightarrow$ $\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$ $\rightarrow$ condition for transportation problem to have a feasible soln.

Problem satisfying this condition are called balanced transportation problems

$\Rightarrow$ A feasible soln. to a transportation problem is said to be a basic feasible soln. if it contain at the most $(m+n-1)$ strictly +ve allocation; otherwise soln. will degenerate.

If total no. of +ve allocation is exactly $(m+n-1)$ then basic feasible soln. is said to be non-degenerate

$\Rightarrow$ feasible soln which minimizes transportation cost is called an optimal soln.

m-origin
n-destination

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L19 - 5a

Origin	Destination 1	2	3	j		n	Supply
1	c_{11}	c_{12}	c_{13}	c_{1j}		c_{1n}	a_1
2	c_{21}	c_{22}	c_{23}	c_{2j}		c_{2n}	a_2
							$\vdots$
i	c_{i1}	c_{i2}	c_{i3}	c_{ij}		c_{in}	a_i
m	c_{m1}	c_{m2}	c_{m3}	c_{mj}		c_{mn}	a_m
Demand	b_1	b_2	b_3	$\cdots$	b_j	$\cdots$ b_n	

$\sum a_i = \sum b_j$

total supply — total demand

mn squares are called cells

Per unit cost c_{ij} of transportation from i^{th} origin to j^{th} destination.

Various a's & b's are called rim requirement

Feasibility of soln. can be verified by summing the value of x_{ij} along rows & down the column

→ As it is L.P.P, one can solve by simplex method

The no. of variables is large & there will be too many steps

As the coefficients of all x_{ij} in constraints being unity, one can go for simplex method or technique.

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Procedure

Step ① Construct transportation table.

Step ② Find initial basic feasible soln.

There are several methods for initial allocation.

North-West corner rule;

Row minima method

Least cost method

Vogel's approximation method (VAM)

↳ takes into account not only least cost C_{ij} but also the cost that just exceed the least cost C_{ij} & ∴ yields a better initial soln.

① Display difference b/w least & next to least cost in each row; by enclosing them in brackets to the right of the row.

Similarly display differences for each column within brackets below that column. ~~In case of~~ ~~the allocation to cell associated with lower~~

② Identify the row or column with the largest difference among all rows & columns & allocate as much as possible under rim requirement, to the lowest cost cell in that row or column. In case of a tie allocate to the cell associated with lower cost

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L19-. 6a

If greatest diff. corresponds to i^{th} row + C_{ij} is the lowest cost in i^{th} row, allocate as much as possible i.e. min (a_i, b_j) in $(i,j)^{th}$ cell + cross off i^{th} row or j^{th} column.

(3) Recalculate row + column differences for the reduced table + go to previous step

(4) Repeat the procedure till all the rim requirement are satisfied.

Step (3) Apply optimality check.

No. of allocations must be '$m+n-1$' otherwise basic soln. degenerate.

To test for optimality, one apply the modified distribution (MODI) method + examine each unoccupied cell to determine whether making an allocation in it reduces total transportation cost + then repeat this procedure until lowest possible transportation cost is obtained

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i) Note no. u_i along left + v_j along top of cost m/x such that their sums equal to original cost of occupied cel.

i.e. solve $u_i + v_j = c_{ij}$

initially start with some $u_i = 0$

ii) Compute net evaluation

$w_{ij} = u_i + v_j - c_{ij}$ for all empty cells.

+ enter them in upper right hand corner of corresponding cells

iii) Examine the sign of each w_{ij}

If all $w_{ij} \leq 0$ then current basic feasible soln is optimal.

If even one $w_{ij} > 0$, soln is not optimal + we proceed

Step 4

① Choose an occupied cell with largest w_{ij} + mark θ in it.

② Draw closed path consisting of horizontal + vertical lines beginning + ending at θ-cell + having it other corner as allocated cell

③ Add + subtract θ alternately to + from the transition cells of the loop subject to sim requirements. Assign a max. value to θ so that one basic variable become zero + other basic variable remain non-negative

Now basic cell whose allocation has been reduced to zero leaves the basis

Step 5 → Return to step 3 + repeat

L19-7a

Soln

		Destination				Availability
		A	B	C	D	
Source	I	21	16	25	13	11
	II	17	18	14	23	13
	III	32	27	18	41	19
Requirement		6	10	12	15	43

Total availability & total requirement is same : 43

Problem is balanced

Find initial basic feasible soln.

Following VAM

Vogel's Approx. Method

21	16	25	11	11(3)	16-13
17	18	14	23	13(3)	17-14
32	27	18	41	19(9)	27-18
6	10	12	15		
(4)	(2)	(4)	(10)		
21-17	18-16	18-14	23-13		

As adding 11 → row become 0 → 1st row is not used now

largest diff is 10 ↓ associated with 4th column

Since $c_{14}(=13)$ is min cost, we allocate

$$\boxed{x_{14} = \min(11, 15)}$$

This exhausts availability of first row, we remove it

as we look for min requirement or availability

17	18	14	23 (4)	13 (3) → 17-14
32	27	18	41	19 (9) – 20-18
6	10	12	4	
(15)	(9)	(4)	(18)	

↑ largest 18 → 23 lowest [min (4, 3)]

as 11 has gone

15-11

Largest is 18 4th column

Since C_{14} (=23) is max cost, we allocate

$$\boxed{x_{14} = \min(13, 4) = 4}$$

as 4th column is zero remove it

We cross fourth column availability

13-4 → as we delivered 4

6	17	18	14	9 (3)
	32	27	18	19 (9)
	6	10	12	
	(15)	(9)	(4)	

↑

max. 1st column; lowest 17 → min (6, 9)

as C_1 goes zero remove C_1

remove 1st column

9-6 → as 6 delivered

min(10,3)

3	18	14	3 (4)
	27	18	19 (9)
	10	12	
	(9)	(4)	

↑

max first column

lowest 18 → min (10, 3)

R_1 goes zero remove R_1

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L19-8a | min (7,9) min (7,12)

[7] 27	[12] 18	19
7	12	

10-3

Initial basic feasible soln.

21	16	25	[11] 13
[6] 17	[3] 18	14	[4] 23
32	[7] 27	[12] 18	41

$u_i \backslash v_j$	v_1	v_2	v_3	v_4
u_1	21	16	25	[11] 13
u_2	[6] 17	[3] 18	14	[4] 23
u_3	32	[7] 27	[12] 18	41

Step 3 Apply optimality check

No of allocation = $m+n-1$ (i.e. 6), one can apply MODI method (Modified distribution)

i) $u_2 + v_1 = 17$, $u_2 + v_2 = 18$; $u_3 + v_2 = 23$

$u_3 + v_3 = 18$, $u_1 + v_4 = 13$; $u_2 + v_4 = 23$

Let $u_2 = 0$ then $v_1 = 17$, $v_2 = 18$, $u_3 = 9$

$v_3 = 9$ $v_4 = 23$, $u_1 = -10$

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② Net evaluation

$w_{ij} = (u_i + v_j) - c_{ij}$ for all empty cells are.

$w_{11} = -10 + 17 - 21 = -14$

$w_{12} = -10 + 18 - 16 = -8$

$w_{13} = -10 + 9 - 25 = -26$

$w_{23} = 0 + 9 - 14 = -5$

$w_{31} = 9 + 17 - 32 = -6$

$w_{34} = 9 + 23 - 41 = -9$

$u_i \backslash v_j$	17	18	9	23
-10	– / 21	– / 16	– / 25	11 / 13
0	6 / 17	3 / 18	– / 14	4 / 23
9	– / 32	7 / 27	12 / 18	– / 41

Since all net evaluations are – ve, current soln is optimal

$x_{14} = 11 \times 13 + 6 \times 17 + 3 \times 18 + 4 \times 23 + 7 \times 27 + 12 \times 18 = 796$

Ex

	Project site 1	2	3	4	Supply
Factories 1	2	3	11	7	6
2	1	0	6	1	19
3	5	8	15	9	10
Demand	7	5	3	2	

to are portation cost per unit from each factories

L19-9a

	1	2	3	4	Supply
1	2	3	11	7	6
2	1	0	6	1	1
3	5	8	15	9	10
Demand	7	5	3	2	17

Following VAM Vogel

diff b/w smallest + next to smallest each row + each column

	2	3	11	7	6 (1)
X →	1	0	6	1	1 (1)
	5	8	15	9	10 (3)
	7	5	3	2	17
	(1)	(3)	(5)	(6)	

$R_2 \to 0$

largest is 4th colum · 6

since $C_{24}(=1)$ is the min. cost, we allocate $x_{24} = \min(1, 2) = 1$

as row is min. remove row ($C_2 \to 0$)

2	5	3	11	7	6 (1)
5		8	15	9	10 (3)
7		5	3	2-1=1	
(3)		(5)	(4)	(2)	

highest 2nd lowest is colum 1st row (3)

$\min(5, 6) = 5$

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L19-9b.

remove 2nd column

3ev

[1] 2	11	7	6-5 = 1 (5) ← highest
5	15	9	10 (4)
7	3	1	
(3)	(4)	(2)	

~~lowest 2 → min~~ (

lowest 2 → min (1, 7)
→ 1

as row min, we remove (row) $R_1 \to 0$

[6] 5	[3] 15	[1] 9	10
7-1 = 6	3	1	

	v_1	v_2	v_3	v_4
u_1	[1] 2	[5] 3	11	7
u_2	1	0	6	[1] 1
u_3	[6] 5	8	[3] 15	[1] 9

Using VAM, initial basic feasible soln is
The transportation cost according to this route is

$1\times2 + 5\times3 + 1\times1 + 6\times5 + 3\times15 + 1\times9$

$= 102.$

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L19-ka

Apply optimality check

$(m+n-1) \rightarrow$ no. of allocation

$3+4-1 = 6$

One can apply MODI method (modified distribution)

net evaluation

$w_{ij} = (u_i + v_j) - C_{ij}$

1) $u_1 + v_1 = 2$

$u_1 + v_2 = 3$

$u_2 + v_4 = 1$

$u_3 + v_1 = 5$

$u_3 + v_3 = 15$

$v_3 + v_4 = 9$

$u_1 = 0$; $v_1 = 2$; $v_2 = 3$

$u_2 = -5$

$u_3 = 3$; $v_3 = 12$; $v_4 = 6$

$w_{13} = u_1 + v_3 - C_{13}$

$= 0 + 12 - 11 = +1$

$w_{14} = u_1 + v_4 - C_{14} = 0 + (-3) - 7 = -10$

$u_i \backslash v_j$	2	3	12	6
0	1 2	5 3	(+) 11	(−) 7
−5	(−) 1	(−) 0	(+) 6	1
3	6 5	(−) 8	3 15	9

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Since net evaluations in two cells are +ve, a better soln. need to be found

Step 7 Next iteration

Choose ~~an~~ unoccupied cell with max w_{ij}.
In case of tie select one with lower original cost

- cell (1,3) & (2,3) each have $w_{ij} = 1$.
& out of it (2,3 has lower original cost 6.
∴ write θ in it.

[1] 2	[5] 3	11	7
1	0	θ 6	[1] − θ 1
[6] 5	8	[3] − θ 15	[7] + θ 9

Draw closed path beginning and ending at θ-cell

add/subtract θ alternatively
Assign max. value to θ so that one basic variable become zero + other remain ≥ 0

[1] 2	[5] 3	11	7
1	0	[θ=1] 6	[1−1] 1
5	8	[3−1] 15	[1+1] 9

θ = 1 as all are +ve
if θ = 3 we get −ve value

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L19-11a | Total transportation cost

$1\times2 + 5\times3 + 1\times6 + 6\times5 + 2\times15 + 2\times9 = 101$

b) optimality check $m+n-1=6$, we can apply MOD

1 / 2	5 / 3	+ / 11	− / 7
− / 1	− / 0	1 / 6	− / 1
6 / 5	− / 8	2 / 15	2 / 9

1 − θ / 2	5 / 3	θ / 11	7
1 / 1	0	1 / 6	1
6 + θ / 5	8	2 − θ / 15	2 / 9

try to pick adjacent −θ

θ = 1

0 (−) / 2	5 / 3	1 / 11	(−) / 7
(−) / 1	(−) / 0	1 / 6	(−) / 1
7 / 5	(−) / 8	1 / 15	2 / 9

$5\times3 + 1\times11 + 1\times6 + 7\times5 + 1\times15 + 2\times9$

$= 100$

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Degeneracy in transportation problem

When the no. of basic cells in $m \times n$ transportation table is less than $(m+n-1)$ basic soln degenerates

To remove degeneracy, we assign a small +ve value ε to as many zero-valued variables as need to complete $(m+n-1)$.

ε' are kept till optimum soln is attained

Then $\varepsilon \to 0$

To

From							Supply
	9	12	9	6	9	10	5
	7	3	7	7	5	5	6
	6	5	9	11	3	11	2
	6	8	11	2	2	10	9
Demand	4	4	6	2	4	2	

using VAM

9	12	[5] 9	6	9	10	5
7	[4] 3	7	7	[2] 5	5	$6+\varepsilon=6$
[1] 6	5	[1] 9	11	3	11	2
[3] 6	8	[2] 11	[4] 2	2	10	9
4	4	$6+\varepsilon=6$	2	4	2	

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L 19.12 a

Solve $w_{ij} = (u_i + v_j) - a_{ij}$ $\quad w_{ij} \leq 0$

$u_i \backslash v_j$	4	3	7	0	0	5
2	– / 9	– / 12	[5] / 9	– / 6	– / 9	– / 10
0	– / 7	[4] / 3	[ε] / 7	– / 7	– / 5	[2] / 5
2	[1] / 6	(0) / 5	[1] / 9	(–) / 11	– / 3	– / 11
2	[3] / 6	– / 8	– / 11	[2] / 2	[4] / 2	– / 10

Min transportation cost

$5\times9 + 4\times3 + \varepsilon\times7 + 2\times5 + 1\times6 + 1\times9 + 3\times6 + 2\times2 + 4\times2 = 112 + 7\varepsilon$

112 as $\varepsilon \to 0$

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